

A380X flyPad failures

The A380X flyPad failure list says what each failure really fails, lists the landing gear sensor and actuator failures, and adds brake, trim air, feed tank pump and crossfeed valve failures

Aircraft	A380X
Where to find it	flyPad > Failures (ATA 21, 24, 26, 28, 32, 34); overhead FUEL panel; E/WD, STATUS and SD FUEL page
Branch	core/failures-a380x, based on master (FlyByWire master 255546cb0 merged in); independent of the other core branches.
Status	Gear failures tested in the sim by the user: OK (2026-10-03). Brake / trim air failures and feed pump / crossfeed valve failures tested in the sim by the user: OK (2026-10-04). Clearer names: not sim-tested (labels only, checked by the a380.spec.ts unit tests).
Pictures	No pictures: the failures page and the ECAM alerts are the stock FBW displays.

1. Landing gear sensor and actuator failures listed, ROLLOUT removed
2. Clearer failure names
3. Brake and trim air failures
4. Feed tank pump and crossfeed valve failures

1. Landing gear sensor and actuator failures listed, ROLLOUT removed

Aircraft:	A380X		Where:	flyPad > Failures > ATA 32 Landing Gear		Status:	Tested in the sim by the user: OK (2026-10-03). Unit tests: a380.spec.ts.
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What it is for

The landing gear proximity sensor and jammed actuator failures (ids 32004 to 32025) were already simulated by the common FBW gear system and mapped in the A380X systems, but the flyPad listed only one of them, and with the wrong name: "uplock nose gear #1" failed the **right main gear** uplock sensor #1. All 18 failures are now listed, with the ids and names of the Rust failure map (the same as on the A32NX).

The **ROLLOUT** failure (ATA 22) only set an FCDC input that nothing reads (there is no steering control system yet). It is no longer listed; its id stays in the code so the C++ failure list keeps in step.

What changed in the list

Id	Before	Now
32004	not listed	Proximity sensor damage uplock nose gear #1
32005	not listed	Proximity sensor damage downlock nose gear #2
32006	Proximity sensor damage uplock nose gear #1 (but failed the right gear)	Proximity sensor damage uplock right gear #1
32007	not listed	Proximity sensor damage downlock right gear #2
32008	not listed	Proximity sensor damage uplock left gear #2
32009	not listed	Proximity sensor damage downlock left gear #1
32010 to 32015	not listed	Proximity sensor damage closed / opened nose, right, left gear door #1 / #2 (six entries)
32020	not listed	Nose gear jammed actuator
32021, 32022	not listed	Left / Right main gears (body + wing) jammed actuator

32023	not listed	Nose gear door jammed actuator
32024, 32025	not listed	Left / Right main gear doors jammed actuator
22001	ROLLOUT (ATA 22)	not listed

NOTE

A door *uplock* sensor ("closed ... door") detects the door closed, a door *downlock* sensor ("opened ... door") the door open. The 32021 / 32022 and 32024 / 32025 names come from the clearer names feature below: one failure jams the whole left or right main gear group.

What you should see

Situation	What you should see
A jammed gear actuator failure set, gear lever moved	That gear (or gear door) group stays where it was: the common FBW gear system freezes its actuator.
A proximity sensor failure set	That sensor gives its LGCIS a wrong signal. The gear logic and its indications are the existing FBW gear model: this branch only lists the failures.
ATA 22 list	No ROLLOUT entry.

How to use

1. Open flyPad > Failures > ATA 32 Landing Gear.
2. Pick the sensor or actuator failure by its name (the table above gives what each one fails) and activate it.
3. Deactivate it to clear the failure.

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 32 Landing Gear.	The 12 "Proximity sensor damage ..." and 6 "... jammed actuator" entries are listed, among them "Proximity sensor damage uplock right gear #1" and "Proximity sensor damage uplock nose gear #1".
2	Open flyPad > Failures > ATA 22 Auto Flight.	No ROLLOUT entry (FMC-A, FMC-B, FMC-C, FCU 1, FCU 2 are still there).
3	In flight, gear down, activate Nose gear jammed actuator , then set the gear lever UP.	The nose gear does not retract; the main gears do.
4	Deactivate the failure.	The nose gear retracts.

2. Clearer failure names

Aircraft: A380X | **Where:** flyPad > Failures | **Status:** Not sim-tested (labels only). Unit tests: a380.spec.ts.

What it is for

The failure names were audited ATA by ATA against what the systems code fails (the Rust failure map, the C++ failure lists and the TypeScript consumers). Names that were misleading or misspelt now say what is failed. Only the labels change: the ids and the failures themselves are the same.

Old name and new name

ATA	Before	Now
21	Foward Cargo Isolation Valve / Foward Cargo Extraction Fan	Forward Cargo Isolation Valve / Forward Cargo Extraction Fan
24	TR ESS	ESS TR
24	TR APU	APU TR
24	AC 247XP	AC EHA (247XP)
24	DC 247PP	DC EHA (247PP)
24	DC 309PP	DC APU (309PP)
26	Engine 1 Loop A ... Engine 4 Loop B	Engine 1 fire detection loop A ... Engine 4 fire detection loop B
26	APU Loop A / APU Loop B	APU fire detection loop A / APU fire detection loop B
26	Main Landing Gear Bay Loop A / B	MLG bay fire detection loop A / B
32	LGCIU 1 Power supply, LGCIU 2 Power supply	LGCIS 1 power supply, LGCIS 2 power supply
32	LGCIU 1 Internal error, LGCIU 2 Internal error	LGCIS 1 internal error, LGCIS 2 internal error
32	Main left / right gear jammed actuator	Left / Right main gears (body + wing) jammed actuator
32	Main left / right gear door jammed actuator	Left / Right main gear doors jammed actuator
34	RA SYS A Interrupted (B, C)	RA SYS A antenna interrupted (B, C)
34	RA SYS A Direct Coupling (B, C)	RA SYS A antenna direct coupling (B, C)
34	TERR 1 / TERR 2	TERR SYS 1 / TERR SYS 2

Limitations and why

Limitation	Why	What to do
"AC ESS" (24105) and "AC EMER" (24104) keep their names.	The A380X electrical model calls the real AC ESS busbar "AC ESS SHED" and the AC EMER busbar "AC ESS"; the labels already follow the real busbars.	Use "AC ESS" for the AC ESS busbar, "AC EMER" for the AC EMER busbar.
"XPDR 2" (34004) and "NSS AVNCS ANSU 2" (46002) keep their names but fail nothing on this branch.	Nothing reads them on master. They are wired on <code>core/a380x-elec</code> .	Test them with that branch.

How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 24 Electrical Power.	The entries read "ESS TR", "APU TR", "AC EHA (247XP)", "DC EHA (247PP)", "DC APU (309PP)".
2	Open ATA 26 Fire Protection.	The loop failures read "Engine 1 fire detection loop A" ... "MLG bay fire detection loop B".

3	Open ATA 32 Landing Gear and ATA 34 Navigation.	"LGCIS 1 power supply" ... "LGCIS 2 internal error"; "RA SYS A antenna interrupted", "RA SYS A antenna direct coupling", "TERR SYS 1", "TERR SYS 2".
4	Open ATA 21 Air Conditioning.	"Forward Cargo Isolation Valve" and "Forward Cargo Extraction Fan" are spelt right.
5	Activate "ESS TR" with the aircraft powered.	The ESS TR fails, as "TR ESS" did before (same id 24002).

3. Brake and trim air failures

Aircraft: A380X | **Where:** flyPad > Failures > ATA 32 and ATA 21 | **Status:** Tested in the sim by the user: OK (2026-10-04); the brake leak works but shows nothing on the SD HYD page.

What it is for

The A380X hydraulic and air conditioning models already had brake leak and trim air faults, but the flyPad had no failure for them. Eleven failures are now listed (same ids as on the A32NX for the brakes), and a spec checks that every id of the systems failure map is listed with its label.

flyPad name (ATA)	What it fails
Brakes NORM circuit leak (green hydraulic) (32)	The NORM brakes are fed by the GREEN system in this model: GREEN fluid leaks out (0.1 gal/s) while the circuit is pressurised.
Brakes ALTN circuit leak (yellow hydraulic) (32)	The ALTN brakes (with the brake accumulator) are fed by the YELLOW system: YELLOW fluid leaks out while pressurised.
Brakes ALTN accumulator gas leak (32)	The ALTN brake accumulator loses its gas charge: the stored brake pressure drops when the YELLOW system is not pressurised.
Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo trim air valve jammed (21)	The trim air valve of that zone stays where it was when the failure was set.
Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo duct overheat (hot trim air) (21)	200 deg C air is blown into the zone duct while its hot air valve is open. The zone gets too hot and the controller closes the trim air valves; HOT AIR OFF stops the overheat (A380 FCOM COND DUCT OVHT procedure).

Limitations and why

Limitation	Why	What to do
No ECAM alert and no SD indication for these failures.	The COND DUCT OVHT alert is defined but not raised on this branch, the duct overheat / trim air marks of the SD COND page are not wired, and the SD HYD page has no brake leak indication.	Watch the effect: zone temperatures on the SD COND page, hydraulic quantity / pressure, brake pressure.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Brakes: flyPad > Failures > ATA 32 Landing Gear, the three "Brakes ..." entries.
2. Trim air: flyPad > Failures > ATA 21 Air Conditioning, the "... trim air valve jammed" and "... duct overheat (hot trim air)" entries.

How to test

#	Do this	Expected result
1	A380X on the ground, four engines running (GREEN and YELLOW pressurised). Activate Brakes NORM circuit leak (green hydraulic) .	The GREEN system loses fluid; YELLOW is not affected. Nothing is shown for the leak itself on the SD HYD page.
2	Deactivate it, activate Brakes ALTN circuit leak (yellow hydraulic) .	The YELLOW system loses fluid; GREEN is not affected.
3	Set the parking brake, then stop the engines (YELLOW no longer pressurised). Activate Brakes ALTN accumulator gas leak .	The brake pressure held by the accumulator drops.
4	In flight, HOT AIR pushbuttons ON, activate Cockpit duct overheat (hot trim air) . Open the SD COND page.	The cockpit zone gets hotter; then the trim air valves close. Setting HOT AIR OFF stops the overheat.
5	Activate Cockpit trim air valve jammed , then change the cockpit temperature selection.	The cockpit trim air valve no longer moves: the cockpit temperature stops following the selection; the other zones still regulate.

4. Feed tank pump and crossfeed valve failures

Aircraft: A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL | **Status:** Tested in the sim by the user: OK (2026-10-04), after the fix that holds the MSFS pump / valve switch.

What it is for

Twelve new failures: the main and standby pump of each feed tank and the four crossfeed valves. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, a jammed valve is held where it was. The FAULT lights, the SD FUEL page and the FCOM alerts compare the selection with the real state (A380 FCOM PRO-ABN-ECAM-10-28, DSC-28-20).

flyPad name (ATA 28)	What it fails
Feed tank 1 main pump ... Feed tank 4 main pump	That main pump stops (held off).
Feed tank 1 standby pump ... Feed tank 4 standby pump	That standby pump stops (held off).
Crossfeed valve 1 jammed in place ... Crossfeed valve 4 jammed in place	That crossfeed valve stays where it was.

What you should see

Situation	What you should see
Feed pump failed, its pushbutton ON	FAULT light in the pushbutton; the pump amber on the SD FUEL page.
One main pump failed	FUEL FEED TK 1 MAIN PMP FAULT (tank 1 to 4), SD FUEL page; procedure FEED TK 1 MAIN PMP ... OFF; STATUS INOP SYS FEED TK 1 MAIN PMP.
One standby pump failed	FUEL FEED TK 1 STBY PMP FAULT , crew awareness; STATUS MORE: REDUND LOSS FEED TK 1 STBY PMP.
Main and standby pumps of one tank failed	FUEL FEED TK 1 MAIN + STBY PMPs FAULT (replaces the two single alerts). After takeoff: CROSSFEED 1 ON, CROSSFEED 2 ON (CROSSFEED 3 and 4 for tanks 3 and 4), FEED TK 1 MAIN PMP OFF, FEED TK 1 STBY PMP OFF, AVOID NEG G LOAD; before takeoff only the two pump lines. STATUS: INOP SYS FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY.

Crossfeed valve jammed and its pushbutton moved

The valve disagrees with its pushbutton: **FAULT** light, valve amber on the SD FUEL page, **FUEL CROSSFEED VLV 1 FAULT** (crew awareness), STATUS INOP SYS CROSSFEED VLV 1; if the valve is stuck closed and engine 1 is not running, FEED TK 1 NOT USABLE.

Limitations and why

Limitation	Why	What to do
The alerts are inhibited during the takeoff and the landing.	The FCOM gives the inhibition as a figure: the one of JETTISON VLV NOT CLOSED is used (FWS flight phases 4 to 7, 9 and 10).	Test on the ground before takeoff or in cruise.
A pump turned OFF by the crew also gives the FAULT alert.	FCOM: a pump is faulty when it runs at low pressure, is abnormally not running, or is turned off by the crew (except in flight phases 1 and 12).	Expected behaviour.
The gravity feed ceiling (MAX FL 280, 50/MEA) is not simulated.	No gravity feed model.	None.
A full MSFS restart is needed after an update.	The cockpit behaviour XML changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or crossfeed valve failure.
2. Follow the ECAM procedure with the overhead FUEL panel pushbuttons; check the SD FUEL page.
3. Deactivate the failure to restore the pump or valve.

How to test

#	Do this	Expected result
1	A380X on the ground, engines running, all feed pumps ON, crossfeed valves in their normal position. Open the SD FUEL page.	No FAULT light, no fuel alert.
2	Activate Feed tank 1 main pump .	FAULT light in the FEED TK 1 MAIN pushbutton, pump amber on the SD, E/WD: FUEL FEED TK 1 MAIN PMP FAULT with FEED TK 1 MAIN PMP ... OFF.
3	Set the FEED TK 1 MAIN pushbutton OFF. Open the STATUS page.	The procedure line is done; STATUS INOP SYS: FEED TK 1 MAIN PMP.
4	Set it ON again and also activate Feed tank 1 standby pump .	The single alerts are replaced by FUEL FEED TK 1 MAIN + STBY PMPs FAULT (on the ground: the two pump OFF lines); STATUS: FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY.
5	Deactivate both. Activate Feed tank 2 standby pump only.	FUEL FEED TK 2 STBY PMP FAULT ; STATUS MORE: REDUND LOSS FEED TK 2 STBY PMP.
6	Deactivate it. Activate Crossfeed valve 1 jammed in place , then move the CROSSFEED 1 pushbutton.	The valve does not move: FAULT light, valve amber on the SD, FUEL CROSSFEED VLV 1 FAULT , STATUS INOP SYS CROSSFEED VLV 1.
7	Deactivate the failure.	The valve follows its pushbutton again; the alert clears.